

From PDEs to Chaos: The Lorenz System

PHYS 340 – Methods of Theoretical Physics

Final Lecture

Why End the Course Here?

- Much of theoretical physics begins with a PDE.
- PDEs usually describe systems with infinitely many degrees of freedom.
- But often, the large-scale physics is captured by only a few dominant modes.
- The Lorenz system is a famous example: a fluid PDE reduced to three coupled ODEs.
- The result is not merely simpler mathematics — it reveals chaos.

The Big Question

A natural question for this course is:

How can a field theory in space and time become a finite-dimensional dynamical system?

Today's roadmap:

- 1 Start with thermal convection.
- 2 Expand the fields in spatial modes.
- 3 Truncate to a few dominant amplitudes.
- 4 Obtain a nonlinear system of ODEs.
- 5 Discover that simple deterministic equations can behave chaotically.

PDE viewpoint

- Unknown depends on space and time
- Example: $u(x, z, t)$
- Need boundary conditions
- Usually infinitely many modes

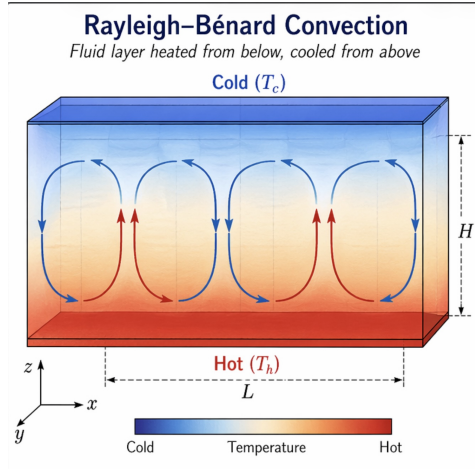
ODE viewpoint

- Unknown depends only on time
- Example: $x(t), y(t), z(t)$
- Finite-dimensional phase space
- Easier to analyze geometrically

The Lorenz model is the bridge between these two viewpoints.

Rayleigh–Bénard Convection

A shallow fluid layer is heated from below and cooled from above. The goal here is to determine: $T(x, z, t)$, $p(x, z, t)$, $\vec{v}(x, z, t)$



Physical Picture of the Instability

- If the temperature difference is small, heat is carried mainly by conduction.
- A fluid parcel that rises enters a region of lower pressure and can remain warmer than its surroundings.
- Warm fluid is less dense, so buoyancy pushes it upward further.
- Likewise, cooler fluid can sink.
- But, **beyond a critical threshold**, the conduction state becomes unstable and convection rolls appear.

This is an instability problem — a central theme of theoretical physics!

Why the Governing Equations Are PDEs

The state of the fluid depends on position and time:

- velocity field $\vec{v}(x, z, t)$
- temperature field $T(x, z, t)$
- pressure field $p(x, z, t)$

So the dynamics involve derivatives such as

$$\frac{\partial T}{\partial t}, \quad \frac{\partial T}{\partial x}, \quad \frac{\partial^2 T}{\partial z^2}, \quad \frac{\partial v_x}{\partial t}, \quad \frac{\partial v_z}{\partial x}, \quad \frac{\partial p}{\partial z}$$

and therefore the equations are PDEs.

We will not derive the full fluid equations today; our focus is the **method of reduction**.

The Mathematical Strategy

This lecture uses a pattern that appears all over PHYS 340:

- 1 Choose basis functions that respect the geometry and boundaries.
- 2 Expand the unknown fields in those basis functions.
- 3 Substitute the expansion into the PDEs.
- 4 Use orthogonality to project onto selected modes.
- 5 Obtain ODEs for the mode amplitudes.
- 6 Truncate to the most important modes.

Expand $\rightarrow$ Project $\rightarrow$ Truncate $\rightarrow$ Analyze

Mode Expansions: The General Idea

Suppose a field depends on space and time:

$$u(x, z, t)$$

Write it as a sum of spatial patterns with time-dependent amplitudes:

$$u(x, z, t) = \sum_{m,n} a_{mn}(t) \phi_{mn}(x, z)$$

Interpretation:

- $\phi_{mn}(x, z)$ describes a spatial shape or pattern.
- $a_{mn}(t)$ tells us how strongly that pattern is excited at time t .

This is the same basic philosophy as Fourier series, normal modes, and eigenfunction expansions.

Why Sines and Cosines?

For rectangular geometry and simple boundary conditions, trigonometric modes are natural:

$$\sin\left(\frac{m\pi x}{L}\right), \quad \cos\left(\frac{m\pi x}{L}\right), \quad \sin\left(\frac{n\pi z}{H}\right)$$

Why they are useful:

- They often satisfy the boundary conditions automatically.
- They are orthogonal on the domain.
- They are ordered from large-scale to fine-scale structure.
- They turn PDEs into algebraic coupling between amplitudes.

Orthogonality Is the Key Step

If the basis functions are orthogonal, then different modes separate cleanly.

Schematically,

$$\int \phi_m(x, z) \phi_n(x, z) dV = 0 \quad (m \neq n)$$

That means when we multiply the PDE by a chosen basis function and integrate, we isolate the equation for that mode.

This is a Galerkin-style projection method.

A Sketch of the Convection Expansion

Edward Lorenz (1963 - Atmospheric Convection) kept only a few carefully chosen large-scale modes. For example, one may write the streamfunction (like electric potential!) and temperature perturbation ($T - T_0$) schematically as

$$\psi(x, z, t) = X(t) \sin\left(\frac{\pi ax}{H}\right) \sin\left(\frac{\pi z}{H}\right)$$

and

$$\theta(x, z, t) = Y(t) \cos\left(\frac{\pi ax}{H}\right) \sin\left(\frac{\pi z}{H}\right) - Z(t) \sin\left(\frac{2\pi z}{H}\right)$$

The exact form is less important today than the idea:

a small number of spatial modes $\Rightarrow$ a small number of amplitudes.

What Is the Streamfunction?

For two-dimensional incompressible flow, write the velocity field as

$$\vec{v}(x, z, t) = (u(x, z, t), w(x, z, t)).$$

Incompressibility ($\vec{\nabla} \cdot \vec{v} = 0$) requires

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0.$$

A convenient way to satisfy this automatically is to introduce the **streamfunction** $\psi(x, z, t)$, defined by

$$u = \frac{\partial \psi}{\partial z}, \quad w = -\frac{\partial \psi}{\partial x}.$$

Then

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = \frac{\partial}{\partial x} \left(\frac{\partial \psi}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \psi}{\partial x} \right) = 0.$$

So ψ encodes the fluid motion, and its amplitude measures the strength of the circulating convection roll.

What Is the Temperature Perturbation?

Write the total temperature field as

$$T(x, z, t) = T_{\text{base}}(z) + \theta(x, z, t).$$

Here:

- $T_{\text{base}}(z)$ is the static conduction profile, depending only on height.
- $\theta(x, z, t)$ is the **temperature perturbation**, the departure from that simple linear profile.

Physically, θ describes the thermal structure created by convection:

- warm fluid rising,
- cool fluid sinking,
- horizontal temperature differences across a convection cell,
- distortion of the vertical temperature gradient.

In the Lorenz truncation:

- $X(t)$ measures the strength of the flow pattern,
- $Y(t)$ measures the horizontal temperature contrast,
- $Z(t)$ measures distortion of the vertical temperature profile.

What Happens Before Truncation?

If we keep *all* modes, then substituting the expansion into the PDEs gives:

- an infinite set of coupled ODEs for the amplitudes
- linear terms from diffusion or damping
- nonlinear coupling terms from advection

So the PDE has not magically disappeared — it has been rewritten as an infinite dynamical system.

If we do not truncate, we are dealing with the full infinite-dimensional fluid PDE rather than a toy model. That places us back in the realm of Navier–Stokes-type dynamics, where some of the deepest open problems in mathematics still live.

The drastic approximation is to keep only the most important amplitudes

Why Truncation Can Work

Why is it even reasonable to keep only a few modes?

- Near the onset of instability, only a few large-scale modes are unstable.
- Higher spatial frequencies are often strongly damped.
- The large-scale qualitative behavior can be controlled by low-order structure.
- A reduced model may capture the essential mechanism even if it misses fine detail.

This is one of the deepest lessons of mathematical physics:

approximation is not the enemy of understanding.

The Lorenz Truncation

Lorenz reduced the dynamics to three time-dependent amplitudes:

$$X(t), \quad Y(t), \quad Z(t)$$

After rescaling variables and parameters, the resulting equations are

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= rx - y - xz, \\ \dot{z} &= xy - bz.\end{aligned}$$

This is the Lorenz system.

How Do We Recover X , Y , and Z ?

The variables X , Y , Z are coefficients in a mode expansion.

If

$$\psi(x, z, t) = X(t)\phi_1(x, z),$$

then orthogonality gives

$$X(t) = \frac{\int \psi(x, z, t) \phi_1(x, z) dV}{\int \phi_1^2(x, z) dV}.$$

Likewise,

$$Y(t) = \frac{\int \theta(x, z, t) \phi_2(x, z) dV}{\int \phi_2^2(x, z) dV}, \quad Z(t) = \frac{\int \theta(x, z, t) \phi_3(x, z) dV}{\int \phi_3^2(x, z) dV}.$$

So X , Y , Z are generalized coordinates obtained by projection onto chosen spatial modes.

What Do the Variables Mean?

The variables are not particle positions. They are mode amplitudes.

- $x(t)$: strength of the convective circulation
- $y(t)$: horizontal temperature contrast
- $z(t)$: vertical distortion of the temperature profile

So the state of the reduced system is a point in a three-dimensional phase space.

Linear Terms and Nonlinear Terms

Look at the structure carefully:

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= rx - y - xz, \\ \dot{z} &= xy - bz.\end{aligned}$$

- Terms like $-x$, $-y$, and $-bz$ act like damping.
- The term rx represents driving controlled by the heating parameter r .
- The products xz and xy are nonlinear couplings.

Those nonlinear couplings are what make the system qualitatively richer than a linear stability problem.

Matrix Form of the Lorenz System

Let

$$\mathbf{u}(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}.$$

Then the Lorenz equations can be written as

$$\dot{\mathbf{u}} = \begin{pmatrix} -\sigma & \sigma & 0 \\ r & -1 & 0 \\ 0 & 0 & -b \end{pmatrix} \mathbf{u} + \begin{pmatrix} 0 \\ -xz \\ xy \end{pmatrix}.$$

So the Lorenz system is:

- a **linear matrix system** controlled by σ , r , and b ,
- plus a **nonlinear coupling term**.

Near an equilibrium point, the linear part is what connects to the eigenvalue/eigenvector analysis we studied earlier.

Equilibria of the Lorenz System

Set

$$\dot{x} = \dot{y} = \dot{z} = 0.$$

One obvious equilibrium is

$$(x, y, z) = (0, 0, 0),$$

which corresponds roughly to no convective circulation.

For sufficiently large r , two additional equilibria appear:

$$(x, y, z) = \left(\pm\sqrt{b(r-1)}, \pm\sqrt{b(r-1)}, r-1 \right).$$

So even the steady-state structure already changes with parameter values.

What Do the Parameters Mean Physically?

Recall the Lorenz equations:

$$\dot{x} = \sigma(y - x),$$

$$\dot{y} = rx - y - xz,$$

$$\dot{z} = xy - bz.$$

The constants σ , r , and b encode the physics of the fluid system:

- r : the **heating strength**

Large r means a larger temperature difference between the hot bottom plate and cold top plate. Increasing r drives stronger convection.

- σ : the ratio of momentum response to thermal response

It compares how quickly the fluid velocity adjusts versus how quickly heat diffuses

- b : a geometric damping factor

It depends on the shape of the convection cells and controls how strongly the z -variable relaxes.

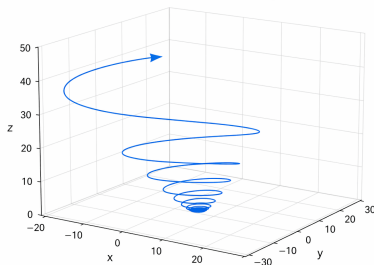
From Conduction to Chaos

Stable Conduction

$$\sigma = 10, \quad r = 0.5, \quad b = \frac{8}{3}$$

Lorenz System: Stable Conduction Regime

$$\sigma = 10, \quad r = 0.5, \quad b = \frac{8}{3}$$

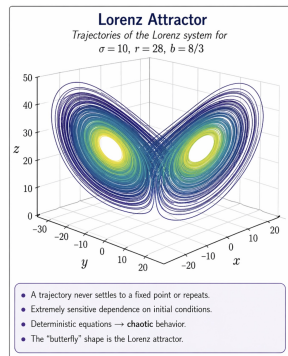


All trajectories decay to the equilibrium at (0, 0, 0).
Conduction state is stable.

Trajectories decay to the equilibrium state.

Chaotic Convection

$$\sigma = 10, \quad r = 28, \quad b = \frac{8}{3}$$



Persistent motion on the strange attractor.

From Stability to Chaos

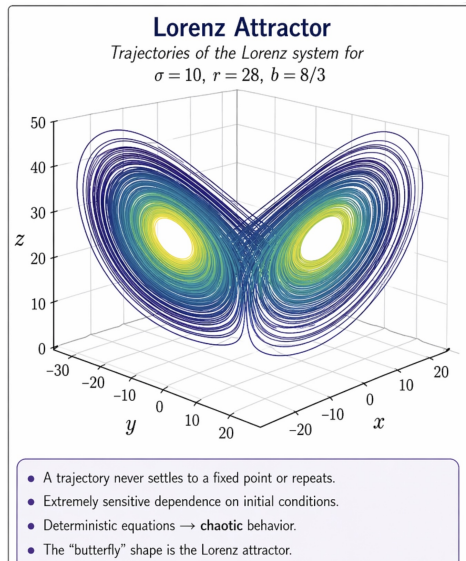
For some parameter ranges, trajectories approach a fixed point.
For the classic parameter set

$$\sigma = 10, \quad r = 28, \quad b = \frac{8}{3}$$

trajectories do something far more interesting:

- they do not settle to a steady state
- they do not approach a simple periodic orbit
- they keep wandering forever in a bounded region of phase space

The Lorenz Attractor



What Makes This So Striking?

This system has only

- three variables
- smooth deterministic equations
- no random forcing

And yet the long-term motion is irregular and effectively unpredictable.

This was historically shocking because it challenged the intuition that deterministic laws automatically imply long-term predictability.

Sensitive Dependence on Initial Conditions

Suppose two initial conditions differ by only a tiny amount.
In a chaotic regime:

- the two trajectories remain close only for a limited time
- then they separate dramatically
- long-term prediction becomes practically impossible

This is the mathematical content behind the phrase *the butterfly effect*.

What the Lorenz Model Does *Not* Mean

Important caution:

- Chaos is **not** the same as randomness.
- The system is still completely deterministic.
- If you know the exact initial condition, the future is uniquely determined.
- The problem is that tiny uncertainty in initial data gets amplified.

So the issue is not absence of law — it is extreme sensitivity within lawful dynamics.

Why This Fits PHYS 340 So Well

This example brings together many methods from the course:

- PDEs and boundary conditions
- separation into spatial modes
- orthogonality and projection
- infinite series expansions
- approximation and truncation
- nonlinear ODE systems
- equilibrium and stability analysis

It is a beautiful capstone example because the mathematics and the physics both matter.

A Broader Moral of Theoretical Physics

The Lorenz system teaches several general lessons:

- ① Very complicated physical systems may have low-dimensional cores.
- ② A crude approximation can still reveal profound truth.
- ③ Simple equations can generate astonishingly rich behavior.
- ④ Determinism does not guarantee predictability.

Ideas growing out of Lorenz's work now appear throughout science:

- weather and climate modeling
- turbulence and fluid mechanics
- nonlinear optics and lasers
- circuits and control systems
- ecology and population dynamics
- celestial mechanics and dynamical astronomy

Chaos theory became a major part of modern applied mathematics and physics.

Expand $\rightarrow$ Project $\rightarrow$ Truncate $\rightarrow$ Analyze

From an infinite-dimensional PDE to a three-dimensional phase portrait.

That is an extraordinary example of how mathematical methods create physical insight.

Sometimes three equations are enough.

And sometimes they are enough to change science.